# Circuit Rewriting Steps: `lemma-S↓HCZH`

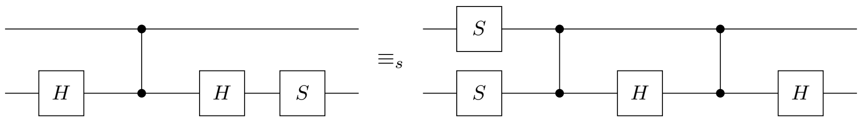
The lemma `lemma-S↓HCZH` in `N/Ex.agda` (line 100) states:

$$S \cdot H \cdot CZ \cdot H \;\equiv_s\; H \cdot CZ \cdot H \cdot CZ \cdot S^{\uparrow} \cdot S$$

where $\cdot$ is matrix multiplication order (rightmost applied first to the state), wire 0 is bottom (↓), wire 1 is top (↑).

## Main theorem

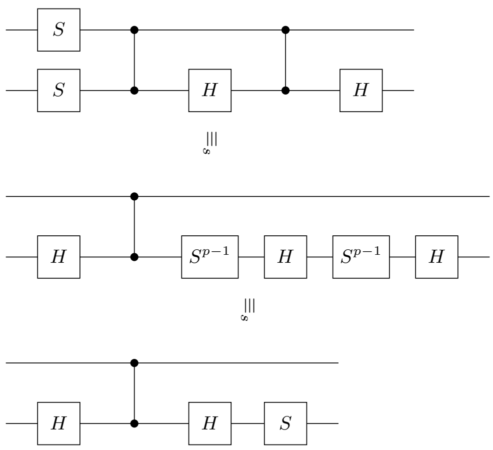
$$S \cdot H \cdot CZ \cdot H \;\equiv_s\; H \cdot CZ \cdot H \cdot CZ \cdot S^{\uparrow} \cdot S$$



## Proof chain

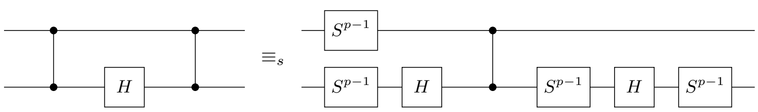
The proof proceeds from the right-hand side $H \cdot CZ \cdot H \cdot CZ \cdot S^{\uparrow} \cdot S$ to the left-hand side $S \cdot H \cdot CZ \cdot H$ in two main steps:

1. Apply **selinger-c11** to the inner $CZ \cdot H \cdot CZ$, then cancel $S^{-1\uparrow} \cdot S^{\uparrow} = \varepsilon$ and $S^{-1} \cdot S = \varepsilon$.
2. Apply the Hadamard conjugation $H \cdot S^{-1} \cdot H \equiv_s S \cdot H \cdot S$ to the leftmost triple, then cancel $S \cdot S^{-1} = \varepsilon$.



## Key lemmas

### selinger-c11 (axiom): $CZ \cdot H \cdot CZ \equiv_s S^{-1} \cdot H \cdot S^{-1} \cdot CZ \cdot H \cdot S^{-1} \cdot S^{-1\uparrow}$



### Hadamard conjugation: $H \cdot S^{-1} \cdot H \equiv_s S \cdot H \cdot S$